



**RV College of
Engineering®**

Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

2025 Scheme

PHYSICS CYCLE

Model Question Papers

RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU)
Department of Artificial Intelligence and Machine Learning
INTRODUCTION TO PYTHON PROGRAMMING
(2025 SCHEME)

Time: 03 Hours**Maximum Marks: 100****Instructions to Candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 and 3 is compulsory. Answer any one full question from 4 or 5, 6 or 7 and 8 or 9.

Part A

Sl. No.		Questions	Marks	BTL	CO
1	1.1	Which data type is used to store a sequence of characters	1	1	1
	1.2	Compare / and // operators in Python.	1	2	1
	1.3	What will be the output of below Python code? str1= "information" print(str1[2:8])	1	3	1
	1.4	How do you create a tuple with only one element?	1	1	2
	1.5	Which method is used to return all keys from a dictionary?	1	1	2
	1.6	Write a function definition to add two numbers and return the result.	1	3	2
	1.7	Give an example of insert() and append() function for the list	1	2	2
	1.8	Write an example for Name Error Exception in Python.	1	3	3
	1.9	Which method in Pandas gives a statistical summary of numerical columns?	1	1	4
	1.10	Which method is used to identify missing values in a DataFrame?	1	1	4

Part B

UNIT - I					
2	a	Explain the different types of conditional statements and how they control the flow of a program. Provide examples to support your explanation.	9	2	2

	b	A software development trainee is building a small Python program that takes user information from the terminal, processes it, and stores it as text output. As part of the task: - The trainee needs to get input (such as name and age) from the terminal. - The trainee must convert the entered age (string input) into an integer to perform arithmetic operations. - Finally, the trainee should create and print a string message combining both values in a readable format. Illustrate how the trainee can implement the above requirements using suitable Python examples for: - Getting input from the terminal - Type casting - Creating strings	9	3	2
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UNIT - II					
3	a	What is a tuple? Explain immutability with an example. List any three built-in tuple methods with examples.	9	2	3
	b	In an engineering materials laboratory, the hardness test results of five metal samples are stored in a Python list: hardness = [45, 55, 60, 50, 48] Perform the following operations using appropriate list methods: - Change the hardness value of the second sample to 58. - Add a new sample hardness value 52 to the list. - Remove the lowest hardness value. - Organize the list in ascending order. - Display the final list after all operations. Also explain how the list data structure helps in solving this engineering data management problem.	9	2	2

UNIT - III					
4	a	Write Python code to perform the following operations on a dictionary: - Add a new key-value pair - Update the value of an existing key - Print all keys and values in the dictionary - Delete a key-value pair	9	3	2
	b	Differentiate between default arguments and keyword arguments? Explain with examples.	9	3	3
OR					
5	a	Explain the use of copy(), get(), has_key() and update() dictionary methods with syntax and examples.	9	2	2

	b	Differentiate between a user-defined function and a built-in function with an example and illustrate how function definition help in avoiding code repetition.	9	3	2
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UNIT – IV					
6	a	Differentiate between static methods and class methods with respect to their behaviour, syntax, and usage. Support your answer with suitable example for both methods.	9	3	2
	b	Explain the concept of encapsulation and its importance in Object-Oriented Programming. Illustrate how data hiding and controlled access are achieved in Python.	9	2	2
OR					
7	a	Illustrate how exception handling in Python helps manage runtime errors and ensures program stability. Support your answer with an example.	9	2	2
	b	A software company is developing a Vehicle Management System. The system needs to handle different types of vehicles such as Car and Bike, which share some common features like start_engine() and display_info(), but each has its own specific behaviour. Task: As a Python developer, you are asked to implement the following: - Create a base class Vehicle with a method display_info() that prints "This is a vehicle". - Derive two subclasses – Car and Bike – from the Vehicle class. - Override the display_info() method in both subclasses to print messages specific to each type of vehicle. - Demonstrate polymorphism by calling the display_info() method using a common reference for different objects.	9	2	2

UNIT – V					
8	a	How do you calculate the mean, median, and standard deviation of a NumPy array with a simple example.	9	3	3
	b	Explain how to customize plot titles, axis labels, and legends.	9	2	2
OR					
9	a	Create and demonstrate Data Frame from a dictionary, list of lists, or a NumPy array with simple examples.	9	3	3
	b	How do you create a NumPy array from a Python list? Provide an example for a 1D and a 2D array.	9	2	2

Signature of the Scrutinizer

Name:

Signature of the BoE Chairperson

Name:

RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU)
Department of Artificial Intelligence and Machine Learning
AI FOUNDATIONS FOR ENGINEERS
(2025 SCHEME)

Time: 03 Hours**Maximum Marks: 100****Instructions to Candidates:**

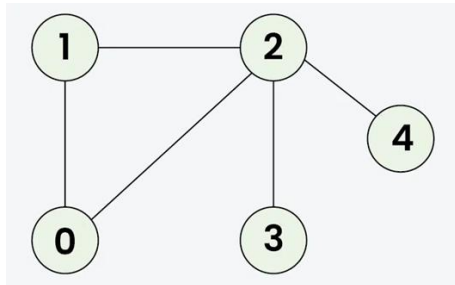
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 and 3 is compulsory. Answer any one full question from 4 or 5, 6 or 7 and 8 or 9.

Part A

1	1.1	A delivery drone automatically adjusts its flight path to avoid obstacles detected by its sensors. Which AI approach does this behaviour represent - acting humanly, thinking rationally, or acting rationally? Justify in one line.	1	L2	CO1
	1.2	An AI-powered hiring system learns from past hiring data that contained hidden gender bias. What step should be taken before retraining the system to make it more responsible?	1	L3	CO4
	1.3	An AI agent navigates a maze where all moves cost the same and the goal is to find the shortest path. Which search strategy - BFS, DFS, or A* - guarantees the shortest path in this scenario?	1	L2	CO2
	1.4	In a self-driving car route planner, the estimated distance to the destination (heuristic) sometimes overestimates the true cost. How does this affect the optimality of the A* algorithm's solution?	1	L4	CO2
	1.5	A spam filter that uses Naïve Bayes misclassifies new types of spam emails not seen in its training data. Which core ML concept does this limitation illustrate - underfitting, overfitting, or poor generalization?	1	L2	CO3
	1.6	An e-commerce recommender uses K-Nearest Neighbour (KNN). When k is increased from 3 to 15, accuracy improves slightly but the model reacts slower to new trends. Explain in one line how increasing k affects model behaviour.	1	L3	CO3
	1.7	Give the purpose of weights in Artificial Neural Networks (ANNs).	1	L1	CO3
	1.8	If the training examples are not linearly separable, which rule is used to converge toward a best-fit approximation to the target concept?	1	L2	CO3
	1.9	Give any one purpose of data preprocessing while building AI models.	1	L1	CO3
	1.10	Define Inclusive AI development.	1	L1	CO4

Part B

UNIT - I					
2	a	1) List and briefly explain the four main approaches to building intelligent systems (4 Marks). 2) List and briefly describe any four foundational disciplines that contributed to the development of Artificial Intelligence (4 Marks).	8	L1	CO1
	b	Design a PEAS model for an AI-enabled irrigation management system that adjusts watering schedules based on soil moisture, rainfall, and crop type. Explain briefly how the system's environment can be described in terms of determinism and observability.	5	L3	CO1
	c	A government agency introduces an AI-based citizen-feedback analyser that categorizes complaints into priority levels. Analyze: 1) Any two possible biases that may appear during training or deployment. (2 Marks) 2) Two mitigation techniques rooted in Responsible AI principles to reduce such bias. (2 Marks) 3) Why accountability is essential when deploying AI for public-service systems. (1 Mark)	5	L3	CO4

UNIT - II																			
3	a	What is a search problem in Artificial Intelligence? List the components required to formulate a search problem. In the Vacuum World problem, what are the two main actions performed by the agent?	8	L1	CO2														
	b	Apply the Depth-First Search (DFS) algorithms on the following graph:  1) Show the order of node visited for DFS starting at node 0. (3 Marks) 2) Discuss one advantage and one limitation of DFS strategy for this graph. (2 Marks)	5	L3	CO2														
	c	An autonomous delivery drone must travel between warehouse A and customer G through the following network. Edge costs represent distance (in km); heuristic values $h(n)$ approximate remaining distance to G. <table border="1" data-bbox="566 1807 909 2002"><thead><tr><th>Edge</th><th>Cost</th><th>$h(n)$</th></tr></thead><tbody><tr><td>A-B</td><td>2</td><td>$h(A)=10$</td></tr><tr><td>A-C</td><td>3</td><td>$h(B)=8$</td></tr><tr><td>B-D</td><td>4</td><td>$h(C)=6$</td></tr><tr><td>C-E</td><td>2</td><td>$h(D)=4$</td></tr></tbody></table>	Edge	Cost	$h(n)$	A-B	2	$h(A)=10$	A-C	3	$h(B)=8$	B-D	4	$h(C)=6$	C-E	2	$h(D)=4$	5	L4
Edge	Cost	$h(n)$																	
A-B	2	$h(A)=10$																	
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B-D	4	$h(C)=6$																	
C-E	2	$h(D)=4$																	

		<table><tr><td>D-F</td><td>3</td><td>$h(E)=3$</td></tr><tr><td>E-F</td><td>2</td><td>$h(F)=2$</td></tr><tr><td>F-G</td><td>2</td><td>$h(G)=0$</td></tr></table>	D-F	3	$h(E)=3$	E-F	2	$h(F)=2$	F-G	2	$h(G)=0$			
D-F	3	$h(E)=3$												
E-F	2	$h(F)=2$												
F-G	2	$h(G)=0$												
		1) Use the A* algorithm to find the optimal path from A to G, showing the $g(n)$, $h(n)$, and $f(n)$ values for each node expanded. (4 Marks) 2) Briefly compare how A* differs from BFS and DFS in terms of efficiency and optimality. (1 Mark)												

UNIT - III																														
3	a	1) What is a well-posed learning problem in Machine Learning? List the three essential elements of a well-posed learning problem (3 Marks). 2) Define the following terms used in designing a learning system (3 Marks): a) Training Experience b) Target Function c) Performance Measure	6	L1	C03																									
	b	The system collects three attributes for each product sample: • Surface Finish (Smooth / Rough) • Weight (Light / Heavy) • Colour (White / Gray) and the target attribute is whether the product is Defective (Yes/No). <table border="1"><thead><tr><th>Sample</th><th>Surface Finish</th><th>Weight</th><th>Color</th><th>Defective</th></tr></thead><tbody><tr><td>1</td><td>Smooth</td><td>Light</td><td>White</td><td>No</td></tr><tr><td>2</td><td>Smooth</td><td>Heavy</td><td>Gray</td><td>No</td></tr><tr><td>3</td><td>Rough</td><td>Light</td><td>Gray</td><td>Yes</td></tr><tr><td>4</td><td>Rough</td><td>Heavy</td><td>White</td><td>Yes</td></tr></tbody></table> 1) Compute the entropy of the target attribute Defective. (2 Marks) 2) Calculate the information gain for the attribute Surface Finish. (4 Marks) 3) Based on your result, comment whether Surface Finish is a suitable root node for the decision tree. (1 Mark)	Sample	Surface Finish	Weight	Color	Defective	1	Smooth	Light	White	No	2	Smooth	Heavy	Gray	No	3	Rough	Light	Gray	Yes	4	Rough	Heavy	White	Yes	7	L3	C03
	Sample	Surface Finish	Weight	Color	Defective																									
1	Smooth	Light	White	No																										
2	Smooth	Heavy	Gray	No																										
3	Rough	Light	Gray	Yes																										
4	Rough	Heavy	White	Yes																										
c	Explain the following aspects related to Decision Tree Learning: 1) How the algorithm handles continuous-valued attributes (e.g., temperature). (2 Marks) 2) How the algorithm prevents overfitting in large datasets. (2 Marks) 3) Two real-world applications of decision tree learning in engineering systems. (1 Mark)	5	L2	C03																										
OR																														
4	a	An online retail company wants to predict whether a visitor will purchase a product (Yes/No) based on browsing behaviour and discounts offered. The system uses a Naïve Bayes Classifier with the following probabilities obtained from training data.	6	L3	C03																									

		<table><tr><th>Class (Purchase)</th><th>P (Class)</th><th>P (Time = Long Class)</th><th>P (Discount = Yes Class)</th></tr><tr><td>Yes</td><td>0.6</td><td>0.5</td><td>0.7</td></tr><tr><td>No</td><td>0.4</td><td>0.4</td><td>0.3</td></tr></table>	Class (Purchase)	P (Class)	P (Time = Long Class)	P (Discount = Yes Class)	Yes	0.6	0.5	0.7	No	0.4	0.4	0.3											
Class (Purchase)	P (Class)	P (Time = Long Class)	P (Discount = Yes Class)																						
Yes	0.6	0.5	0.7																						
No	0.4	0.4	0.3																						
		For a new customer with the following features: - Time on site = Long - Discount = Yes Perform the following: 1) Compute the unnormalized posterior probabilities for each class. (4 Marks) 2) Identify the predicted class for the customer. (2 Marks)																							
b		An e-commerce company uses a K-Nearest Neighbor (KNN) algorithm to predict whether a customer will purchase a product (Yes/No). The model considers two features for each customer: - Time on Site (minutes) - Number of Product Views The training dataset is given below. <table><tr><th>Customer</th><th>Time on Site (min)</th><th>Views</th><th>Purchase</th></tr><tr><td>1</td><td>2</td><td>3</td><td>No</td></tr><tr><td>2</td><td>3</td><td>4</td><td>No</td></tr><tr><td>3</td><td>5</td><td>6</td><td>Yes</td></tr><tr><td>4</td><td>8</td><td>8</td><td>Yes</td></tr></table> A new customer Q (4,5) visits the website. 1) Compute the Euclidean distance between Q (4,5) and each customer in the dataset using the formula: (4 Marks)$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ 2) Identify the three nearest neighbours (k = 3) and determine the majority class. (2 Marks) 3) Write the final predicted class for customer Q and justify your answer in one sentence. (1 Mark)	Customer	Time on Site (min)	Views	Purchase	1	2	3	No	2	3	4	No	3	5	6	Yes	4	8	8	Yes	7	L3	CO3
Customer	Time on Site (min)	Views	Purchase																						
1	2	3	No																						
2	3	4	No																						
3	5	6	Yes																						
4	8	8	Yes																						
c		1) Explain how the choice of k in KNN affects bias and variance. (2 Marks) 2) Compare Naïve Bayes and KNN in terms of: - Training time - Prediction time - Memory usage (3 Marks)	5	L2	CO3																				

UNIT – IV					
6	a	1. With a block diagram, discuss the working of a Perceptron.	5	L2	CO3
		2. State the perceptron training rule, and discuss various terms in the training rule.	5		
	b	1. Give the principle of the working of Convolutional Neural Networks (CNNs)	4	L2	CO3
		2. CNNs are widely used to classify images into various categories. How can this CNN feature be used to develop a signature recognition system?	4		
OR					

7	a	1. Derive the weight update rule for gradient descent $\Delta w_i = \eta \sum_{d \in D} (t_d - o_d) x_{id}$ starting from $\nabla E(\vec{w}) \equiv \left[\frac{\partial E}{\partial w_0}, \frac{\partial E}{\partial w_1}, \dots, \frac{\partial E}{\partial w_n} \right]$ 2. Write the stochastic gradient version of the Backpropagation algorithm for feedforward networks containing two layers of sigmoid units	5	L3, L4	CO3
	b	1. Give the principle of the working of Recurrent Neural Networks (RNNs) 2. Suppose you are designing a behavioural authentication system based on users' typing speed and rhythm. How can RNNs be utilized in this system?	4 4	L2, L3	CO3

UNIT – V					
8	a	1. How are commonly used AI explanation techniques mapped to different stages of the AI model life cycle, including data collection, training, debugging, and deployment? 2. What are the expectations that various stakeholders may have regarding AI explanations in a credit approval use case?	6 4	L2	CO4
	b	Discuss how Privacy and Fairness play an important role in developing responsible AI design.	8	L3	CO4
OR					
9	a	1. A dataset is considered biased when certain elements are heavily weighted compared to others, ultimately skewing the dataset. Discuss any three types of data bias. 2. Mention any FOUR factors to be considered while sourcing the data to build AI systems.	6 4	L1, L2	CO4
	b	Provide one practical example each for Black-box Testing, Behavioural Testing, Fairness Testing, and Adversarial Testing.	8	L3	CO4

Signature of the Scrutinizer

Name:

Signature of the BoE Chairperson

Name:

COURSE CODE- CV112TA
USN-

RV COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

I Semester B.E. January -2026 Examinations

DEPARTMENT OF CIVIL ENGINEERING

COURSE- ENGINEERING MECHANICS

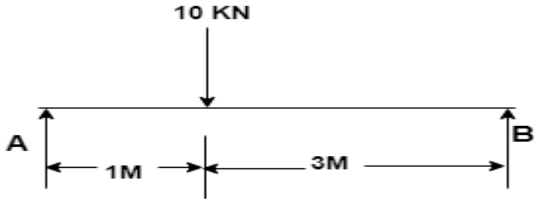
Time-03 Hours

Maximum Marks-100

Instructions to Candidates:

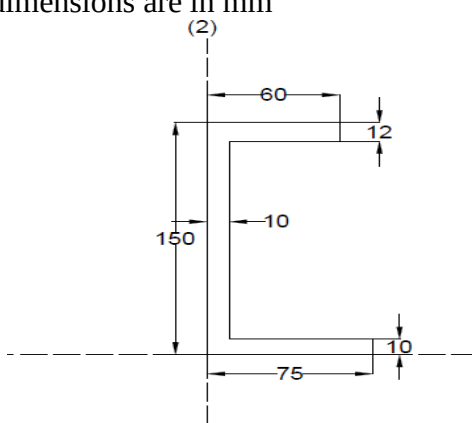
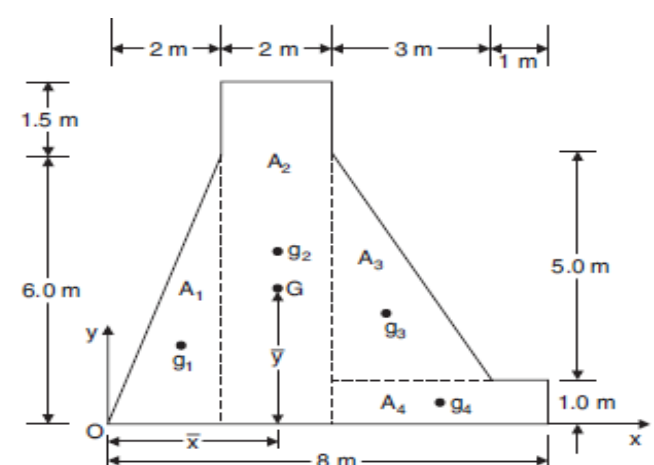
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PART A

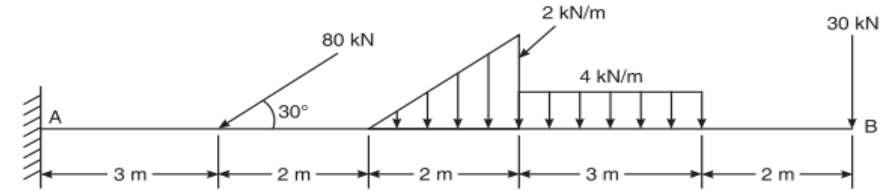
Q No	Questions	Marks
1	1.1 State Perpendicular axis theorem	1
	1.2 Centroid distance of a semicircle of radius 'R' when measured from its circumference is _____	1
	1.3 Find the reaction at the supports for the beam as shown in Fig 1.3  Fig 1.3	2
	1.4 Define 'Point of contra flexure' in a beam	2
	1.5 What is a perfect truss	2
	1.6 State the principle of superposition	2

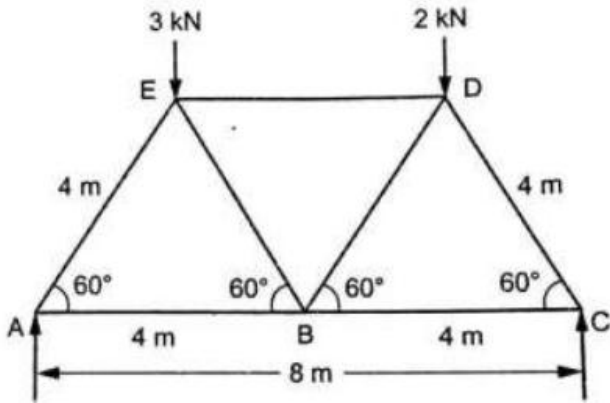
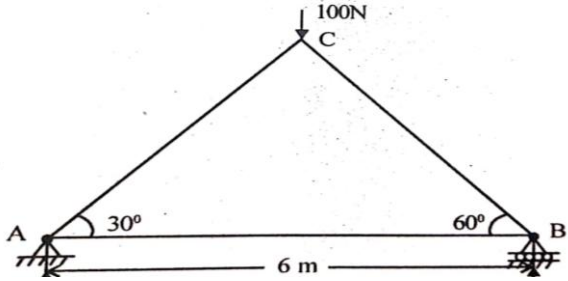
PART B

UNIT-I

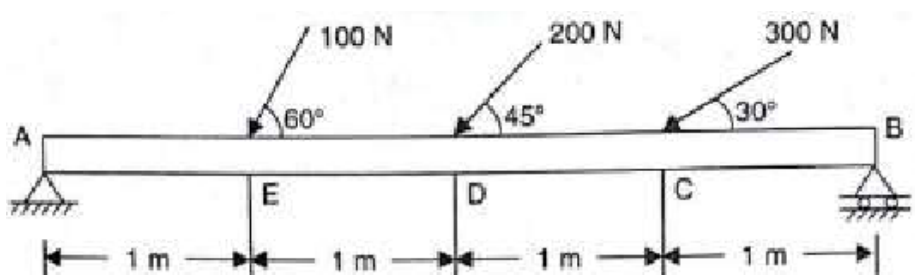
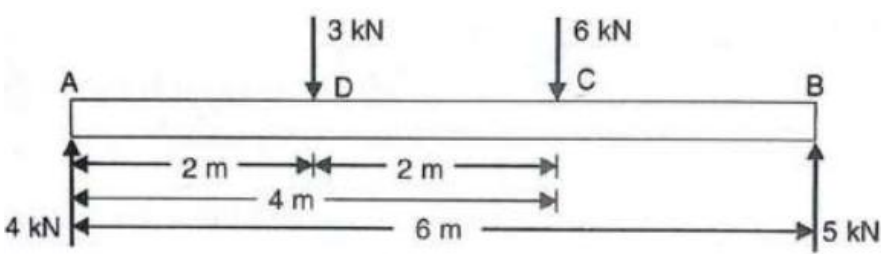
2	a	Derive the centroid coordinates of a Triangle from first principles	5
	b	Find the radius of gyration of the area shown in Fig.2.b about the horizontal centroidal axis. All dimensions are in mm  Fig.2.b	7
	c	Determine the centroid of the section of the concrete dam shown in Fig.2.c  Fig.2.c	7

UNIT-II

3	a	Discuss different types of supports with neat sketches	8
	b	Find the reactions for the cantilever beam loaded as shown in Fig. 3.b  Fig. 3.b	10

UNIT-III			
4	a	Enlist the assumptions made in the analysis of statically determinate trusses	4
	b	A truss of 8m span consisting of seven members each of 4m length supported at its ends and loaded as shown in Fig.4.b. Determine the forces in the members by method of joints  Fig.4.b	14
OR			
5	a	Find the forces in the members AB, BC and AC of the truss shown in Fig.5.a by method of section.  Fig.5.a	18
UNIT-IV			
6	a	With illustrative sketches, discuss the stress strain behavior of Mild steel under tension	12
	b	A material has modulus of rigidity equal to $0.4 \times 10^5 \text{ N/mm}^2$ and bulk modulus equal to $0.8 \times 10^5 \text{ N/mm}^2$. Find the Young's Modulus and Poisson's Ratio	6
OR			
7	a	A specimen of steel 20 mm diameter with a gauge length of 200 mm is tested to destruction. It has an extension of 0.25 mm under a load of 80 kN and the	12

	load at elastic limit is 102 kN. The maximum load is 130 kN. The total extension at fracture is 56 mm and diameter at neck is 15 mm. Find (i) The stress at elastic limit. (ii) Young's modulus. (iii) Percentage elongation. (iv) Percentage reduction in area. (v) Ultimate tensile stress	
b	Derive the relation between Young's modulus and bulk modulus	06

UNIT-V		
8	a	Draw Shear force and bending moment diagram for a cantilever beam with a single Point load at free end
	b	Draw the Shear force and bending moment diagram for the beam shown in Fig.8.b  Fig.8.b
OR		
9	a	Draw the Shear force and bending moment diagram for the beam as shown in Fig.9.a  Fig.9.a
	b	Derive the Relationship between Load, Shear Force and Bending Moment of a beam

Signature of Scrutinizer	Signature of Chairman
Name –Dr. Nethravathi S Dr. Venugopal. G	Name- Dr Anjaneyappa

COURSE CODE- CV113ATB
USN-

RV COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

I Semester BE January -2026 Examinations

DEPARTMENT OF CIVIL ENGINEERING

COURSE- ELEMENTS OF CIVIL ENGINEERING

Time-03 Hours

Maximum Marks-100

Instructions to Candidates:

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PART A

Q No		Questions	Marks
1	1.1	Define Plinth area.	1
	1.2	Name the national green building rating system used in India.	1
	1.3	List various alternative materials used in civil engineering	2
	1.4	Define potable and palatable water.	2
	1.5	Differentiate between shallow and deep foundations	2
	1.6	List the applications of RS and GIS in civil engineering	2

PART B

UNIT-I			
2	a	Describe briefly the scope and significance of Civil Engineering in Structural Engineering	4
	b	Justify the need for multidisciplinary knowledge in Civil Engineering practice.	8
	c	Discuss the role of Civil Engineers in achieving sustainable development goals	6

UNIT-II			
3	a	Explain briefly the engineering properties of bricks.	4
	b	Explain the classification and functional importance of beams and columns in building construction	6
	c	Explain the concept of green buildings and discuss their need in sustainable development	8

UNIT-III			
4	a	Illustrate with a neat diagram the stages involved in a typical water treatment process and describe the functions of each stage	10
	b	Explain how rainwater harvesting systems help reduce dependency on surface water sources and discuss their role in achieving sustainable urban water resource management	8
5	a	Bengaluru city, with a population exceeding 1.4 crore, is presently facing acute challenges in managing municipal solid waste due to rapid urbanization and population growth. The increasing quantity of waste has overburdened collection, segregation, and disposal systems, leading to environmental pollution and health hazards. For the above context, - Identify and classify the major types of solid waste generated in Bengaluru city. - Suggest sustainable practices and innovative solutions that can be adopted to minimize waste generation and enhance recycling efficiency?	10
	b	Discuss briefly the importance of dams, canals, and aqueducts in water resource management	8

UNIT-IV			
6	a	Explain the classification of roads in India	8
	b	Explain the role of public transport systems in reducing traffic congestion and discuss the importance of intermodal connectivity in achieving efficient urban mobility	10
7	a	Explain briefly explain the classification of tunnels	8
	b	Discuss the importance of road safety awareness programs and describe how modern RS and GIS tools assist authorities in planning safer transportation networks	10

UNIT-V			
8	a	Classify different types of foundations and discuss their suitability for various soil conditions	8
	b	Explain how drones and 3D printing are transforming modern construction practices in terms of efficiency and sustainability	10
9	a	During monsoon seasons, your town faces recurrent floods affecting roads and buildings. Analyze how modern tools such as drones, Artificial Intelligence (AI), and sensors can be used to reduce flood risks	10
	b	Discuss the applications of Artificial Intelligence in civil engineering	8

Signature of Scrutinizer	Signature of Chairman
Name –Dr Madhavi K Dr Sindhu D	Name- Dr Anjaneyappa

USN

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RV COLLEGE OF ENGINEERING
Autonomous Institution affiliated to VTU
I Semester B.E. December/January -2025 Examinations
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
BASIC ELECTRONICS
(2025 SCHEME)

*Time: 03 Hours**Maximum Marks: 100***Instructions to candidates:**

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PART-A

1	1.1	The canonical sum of product form of the function $y(A,B) = A + B$ is _____.	1
	1.2	Simplify $AB + BC + B\bar{C}$	1
	1.3	In a voltage divider bias circuit, if $R_1=47\text{ k}\Omega$, $R_2=10\text{ k}\Omega$, and $V_{CC}=12\text{V}$, the base voltage is _____.	1
	1.4	For a BJT in active region, if $V_{BE}=0.7\text{ V}$, $V_{CE}=5\text{ V}$, and $I_C=1\text{mA}$, then the collector resistor R_C is _____.	1
	1.5	An N channel MOSFET with $V_t=1\text{V}$, $K=2\text{mA/V}^2$ has $V_{gs}=V_{ds}=4\text{V}$. The drain current of the transistor is _____.	1
	1.6	The bandwidth of an amplifier with negative feedback increases by a factor of _____.	1
	1.7	The CMRR of an op-amp is 92dB, if its common mode gain is 0.5. Its differential gain is _____ dB.	1
	1.8	In a voltage follower circuit, the op amp used is ideal in all respects except that it has a finite gain of 5000. The gain of the voltage follower is equal to _____.	1
	1.9	The peak value of the carrier signal if the maximum voltage and minimum voltage are 15V and 5V respectively is _____.	1
	1.10	An _____ system is a microcontroller or microprocessor based system which is designed to perform a specific task.	1

PART-B**UNIT-I**

UNIT-I			
2	a	Write the truth table for SUM and CARRYOUT of a full adder. From the truth table, obtain the logic expressions for the same and then realize the full adder using 2 half adders.	06
	b	Simplify the following function $F(A, B, C, D) = \sum m(0,2,3,4,6,7,8) + d(10, 11, 14,15)$ using k-map. Also, realize using only two-input NAND gates.	06

	c	Design a digital circuit to control a street light system that depends on three inputs - Day/Night sensor (A), Traffic sensor (B), and Manual switch (C). The light should turn ON when it is night and traffic is detected, or when the manual switch is ON. Implement this control logic using an 8×1 Multiplexer. Draw the truth table and circuit diagram.	06
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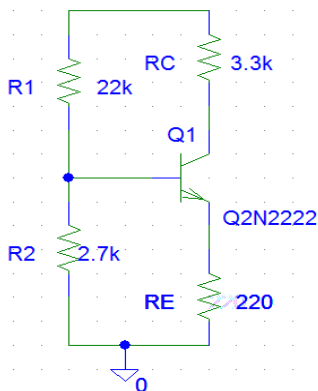
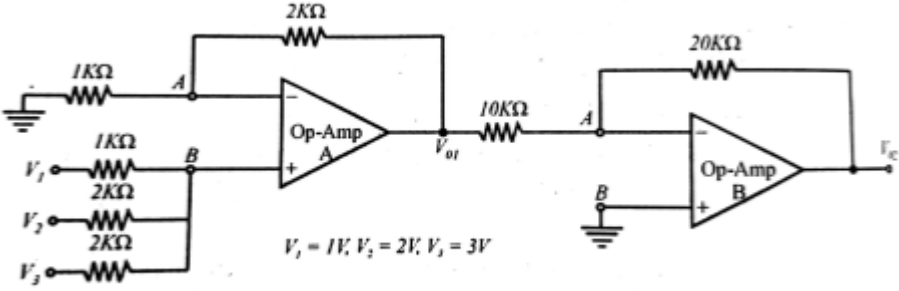
UNIT-II			
3	a	Explain the terms i) Thermal runaway and ii) Early effect, in a BJT	06
	b	Explain how a transistor can be used as a switch.	04
	c	Consider the value of voltage divider bias circuit shown in Fig 3.c, determine the values of V_{CE} and I_C , such that $\beta = 200$, $V_{BE} = 0.7V$, $V_{CC} = 15V$. Draw the Load line. 	08

Fig. 3.c

UNIT-III			
4	a	Draw the structure and explain the operation of an N-channel enhancement type MOSFET. Draw its output characteristics and indicate three regions of operation.	06
	b	An N channel MOSFET having $V_t = 1V$ acts as a resistance of 500Ω for small values of V_{DS} , when $V_{GS} = 3V$. i) If V_{GS} is changed to 6V, for small values of V_{DS} , what is the resistance offered by the MOSFET ii) When $V_{GS} = V_{DS} = 4V$, determine the drain current. iii) Determine the trans conductance when $V_{GS} = V_{DS} = 3V$.	08
	c	Draw the circuit of a 2-input CMOS NAND gate and explain its working.	04
OR			
5	a	An amplifier has an open loop gain of 8000, input resistance = $100k\Omega$, output resistance = $10k\Omega$, distortion = 15%, gain stability = 20%, bandwidth = 500kHz and noise = $1\mu V$. By giving suitable negative feedback, the resistance needs to be increased to $2.2M\Omega$. Determine the feedback factor required. Also determine the gain, the output resistance, distortion, gain stability, bandwidth and noise of the closed loop amplifier.	08

	b	An N-channel enhancement type MOSFET with $V_{th}=1V$ conducts a current $I_D=100\mu A$ when $V_{GS}=V_{DS}=1.5V$. Find the value of I_D for $V_{GS}=2.5V$ and $V_{DS}=4V$. Also calculate the value of r_{DS} for small values of V_{DS} , when $V_{GS}=3V$.	06
	c	Prove that the stability of the gain of an amplifier improves with negative feedback by a factor, $(1+A\beta)$, where A is open loop gain of the amplifier and β is the feedback factor.	04

UNIT-IV			
6	a	List at least Eight important characteristics of an ideal op-amp and indicate their typical values for a general purpose commercial op-amp.	06
	b	Draw the circuit of an integrator using an ideal op-amp and prove that it works as an integrator.	06
	c	Draw a circuit and calculate the values of different resistors of a summer circuit using two ideal op-amp to get $v_o = v_1+3v_2+5v_3-7v_4-9v_5-11v_6$, where v_1, v_2, v_3, v_4, v_5 and v_6 are the available inputs.	06
OR			
7	a	Define Slew rate, CMRR, PSRR and input offset voltage with respect to an OPAMP	06
	b	Calculate the output voltage V_O of the circuit shown in fig.7a  $V_1 = 1V, V_2 = 2V, V_3 = 3V$ Fig.7a	06
	c	Explain the working of a Schmitt Trigger circuit with a neat sketch and draw the appropriate waveforms.	06

UNIT-V			
8	a	Define Modulation. What are the different types of modulation? Explain the need for modulation.	06
	b	An amplitude modulated wave is given by $S(t)=25[1+0.4\sin(1570t)]\sin(62.8*10^6t)$. with carrier power as 2KW, Find f_c , f_m , V_c , Bandwidth, modulation index and P_t .	06
	c	List out any 6 salient features of 8051 microcontroller.	06
OR			
9	a	With the help of a block diagram explain the working of a digital communication system.	06

	b	Derive the expression for an AM signal when single tone message signal is modulated with a carrier. Also explain the frequency components present in AM signal.	06
	c	List any 6 differences between RISC and CISC architecture.	06

Signature of Scrutinizer:

Signature of Chairman

Name:

Name:

Course Code:EC112TA.

Course Title: Basic Electronics

PART-A

Q.No	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
B T	1	2	3	3	4	2	3	1	3	2
COs	1	2	3	4	4	4	4	3	2	1

PART-B

Question No		B T Levels	COs addressed		Question No		BT Levels	COs addressed
2	a	2	1		3	a	1	2
	b	3	1			b	1	2
	c	4	1			c	4	2
4	a	2	3		5	a	3	3
	b	3	3			b	2	3
	c	1	3			c	1	3
6	a	1	4		7	a	1	4
	b	2	4			b	3	4
	c	3	4			c	2	4
8	a	1	1		9	a	1	1
	b	3	2			b	3	2
	c	1	3			c	1	3
10	a				11	a		
	b					b		

Signature of Scrutinizer:

Signature of Chairman

Name:

Name:

USN

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RV COLLEGE OF ENGINEERING
(Autonomous Institution affiliated to VTU)
I Semester B.E. December/January -2025 Examinations
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
PRINCIPLES OF ELECTRONICS ENGINEERING
(2025 SCHEME)

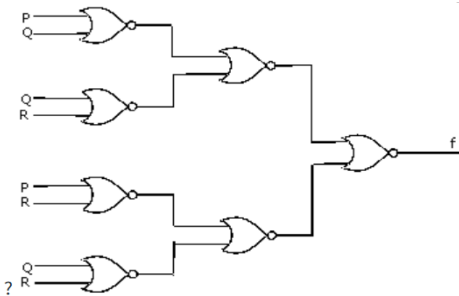
Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

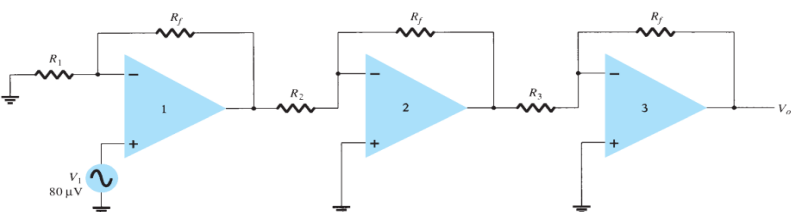
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 and is compulsory. Answer any one full question from 5 or 6, 7 or 8, and 9 or 10.

PART-A

1	1.1	Determine the current through a PN junction diode at 37° C, If the forward voltage drop across the diode is 650mV and its reverse saturation current is 8nA. Assume $\eta=2$.	2
	1.2	A diode has a forward current of 0.5mA and its forward voltage is 500mV. Determine its reverse saturation current. Assume $\eta=2$.	2
	1.3	Photodiodes are operated in _____ bias to detect light efficiently.	
	1.4	How many cells are present in a 3-variable K-map?	1
	1.5	If the differential gain of an amplifier is $A_d = 100,000$ and the CMRR is 80dB, find the common-mode gain A_{cm} .	1
	1.6	A non-inverting amplifier uses resistors $R_1 = 1k\Omega$ and $R_f = 9k\Omega$. Calculate the voltage gain (A_v) of the amplifier.	2
	1.7	What is the Boolean expression for the output f of the combinational logic circuit of NOR gates shown in figure 1.7?  Figure 1.7	1
	1.8	How many AND, OR and EXOR gates are required for the configuration of full adder?	1

PART-B**UNIT-I**

2	a	Write the Shockley's diode current equation and plot the diode characteristics. Explain each parameter of equation and V-I curve.	06
	b	Explain the working principles and applications of Photo diodes and LEDs.	06
	c	Draw the block diagram of a DC power supply and explain the function of each block.	06

UNIT-II			
3	a	With neat diagrams, explain the operation of the RC Phase Shift Oscillator. Derive the expression for frequency of oscillation and explain how the phase shift of 180° is achieved.	06
	b	State and explain the Barkhausen Criterion for sustained oscillations. Why is it necessary for oscillator design?	04
	c	Calculate the output voltage using the circuit of Fig. 3c for resistor components of value $R_f = 470\text{ k}\Omega$, $R_1 = 4.3\text{ k}\Omega$, $R_2 = 33\text{ k}\Omega$, and $R_3 = 33\text{ k}\Omega$ for an input of $80\mu\text{V}$.  Fig. 3c	08

UNIT-III			
4	a	Using Boolean algebra theorems and properties, simplify the following Boolean expression step by step: $F(A, B, C, D) = \overline{A}BC + A\overline{B}D + ABC + AB\overline{C}D$ Show each simplification step clearly and obtain the minimal SOP expression. Also draw the logic circuit using basic gates (AND, OR, NOT) for the simplified result.	06
	b	Design a Half adder and Full Adder circuit and Elaborate its working and give its (i) Circuit Diagram, (ii) Truth Table	08
	c	Explain the Canonical and Standard Forms of Boolean functions with suitable examples. Clearly differentiate between Canonical Sum of Minterms (SOP) and Canonical Product of Maxterms (POS)	04
OR			
5	a	Design a Full adder using two half adders and also derive the expression for sum and carry, along with its circuit diagram.	08
	b	Solve the following Expression Using K Map $F(A, B, C, D) = \sum M(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$	06
	c	State and explain DeMorgan's Laws using Boolean expressions.	04

UNIT-IV			
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6	a	Define amplitude modulation. Describe its process and write the mathematical expression for its waveform. Derive the expression for the total transmitted power of an AM signal in terms of carrier power and modulation index.	06
	b	Differentiate between Micro Processor and Micro controller	06
	c	Elaborate and explain the Von Neumann and Harvard architectures with suitable block diagram.	06
OR			
7	a	With a neat block diagram of Digital Communication system explain the working of each block.	06
	b	An audio frequency signal $10\sin 2\pi \times 500t$ is used to amplitude modulate a carrier of $50\sin 2\pi \times 10^5t$. Calculate (i) Modulation Index , (ii) Side Band Frequencies (iii) Amplitude of Each Side band frequencies , (iv) Bandwidth required (v) Total Power Delivered	06
	c	Differentiate between the RISC and CISC Architecture	06

UNIT-V			
8	a	Explain the key characteristics of embedded systems and classify them based on performance and complexity. Discuss one example from each class with its typical applications.	06
	b	Describe the major design metrics (Power, Size, Cost, Performance) and constraints in embedded systems development. Illustrate with a real-world trade-off scenario, such as in battery-powered devices.	06
	c	Outline the basics of Embedded Linux, including its kernel features, bootloader process, and filesystem support. Compare its advantages over bare-metal programming for IoT applications.	06
OR			
9	a	Define Edge Computing, explaining its need, typical architecture (including fog and cloud layers), and provide two practical examples. Highlight how it addresses latency issues in traditional cloud computing.	08
	b	Describe the steps to build an IoT system using ESP32, integrating cloud platforms like ThingSpeak or Blynk for data visualization. Illustrate with a case study on edge-based predictive maintenance or wearable health monitors.	10

Signature of Scrutinizer:

Signature of Chairman

Name:

Name:

Course Code:EC113ATC.

Course Title: **PRINCIPLES OF ELECTRONICS
ENGINEERING****PART-A**

Q.No	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
B T	3	3	1	2	3	3	3	2		
COs	2	2	1	1	2	3	4	3		

PART-B

Question No		B T Levels	COs addressed		Question No		BT Levels	COs addressed
2	a	2	1		3	a	2	1
	b	2	1			b	1	2
	c	2	1			c	3	3
4	a	4	3		5	a	4	3
	b	3	4			b	3	4
	c	2	3			c	2	2
6	a	2	3		7	a	1	3
	b	2	2			b	3	2
	c	1	2			c	2	1
8	a	1	4		9	a	1	4
	b	2	1			b	2	4
	c	1	4			c		
10	a				11	a		
	b					b		

Signature of Scrutinizer:

Signature of Chairman

Name:

Name:

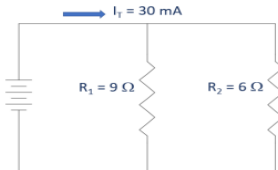
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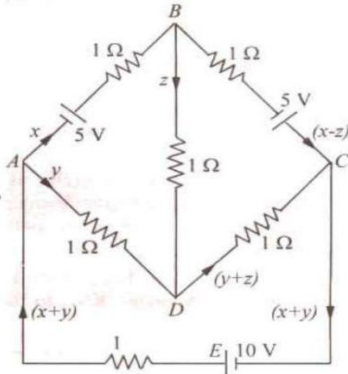
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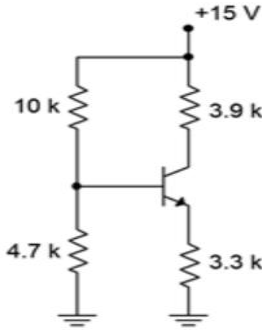
RV COLLEGE OF ENGINEERING
Autonomous Institution affiliated to VTU
ISemester B.E. Jan-2026 Examinations
ELECTRICAL AND ELECTRONICS ENGINEERING
ELEMENTS OF ELECTRICAL ENGINEERING
(2025 SCHEME)

Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

		PART-A (Objective type for one or two marks) (True & false, Fill in the Blanks and match the following questions are not permitted)	
1	1.1	Use the current divider rule to determine the currents flowing through R_1 and R_2 in the circuit shown in Fig.1.1  Fig. 1.1	1
	1.2	What is the difference between statically and dynamically induced emf?	1
	1.3	Two wattmeters are used to measure the power in 3 phase balanced system. What is the power factor when one reads twice the other?	2
	1.4	A geyser is rated at 3kW, 230V, 50Hz. If it is switched ON for one hour daily, what would be the energy cost saving, at the rate of Rs. 2.50 per unit if it is replaced by a solar water heater?	2
	1.5	Plot the VI Characterstics of power diode.	2
	1.6	What is the purpose of earthing electrical appliances?	1
	1.7	Mention the main components of electric vehicles.	1
PART-B (Maximum subdivisions is limited to 3 in each question)			
UNIT-I			
2	a	What is the significance of coupling coefficient? Derive the expression for coefficient of coupling.	6
	b	Determine the branch currents in the network of Fig. 2.b when the value of each branch resistance is one ohm.	8

		 Fig. 2.b																					
	c	State and explain kirchoff's laws.	4																				
UNIT-II																							
3	a	An alternating voltage is represented by the equation $v=100 \sin 520t$ V. Calculate its i) maximum value, ii) rms value, iii) average value, iv) form factor, v) peak factor, vi) frequency, vii) voltage when $t=0.0015s$	6																				
	b	Two wattmeters measure total power in a 3 phase circuit and are correctly connected. One wattmeter reads 4800 W and the other reads backwards. On reversing the connection of the latter, it reads 400 W. What is the total power and power factor.	6																				
	c	With the aid of a phasor diagram, obtain the relationship between the line and phase values of voltage in a three-phase, star connected system.	6																				
UNIT-III																							
4	a	Estimate Total Daily Energy Requirement for the following loads. <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th>Name of the Appliance</th><th>Power Rating (W)</th><th>Avg. Daily Usage Hrs</th><th>No. of Appliances</th></tr> </thead> <tbody> <tr> <td>CFL</td><td>12</td><td>6</td><td>3</td></tr> <tr> <td>Fan</td><td>50</td><td>8</td><td>2</td></tr> <tr> <td>TV (21")</td><td>150</td><td>2</td><td>1</td></tr> <tr> <td>Computer</td><td>250</td><td>3</td><td>1</td></tr> </tbody> </table> Take electricity cost to be Rs.6 per unit.	Name of the Appliance	Power Rating (W)	Avg. Daily Usage Hrs	No. of Appliances	CFL	12	6	3	Fan	50	8	2	TV (21")	150	2	1	Computer	250	3	1	6
Name of the Appliance	Power Rating (W)	Avg. Daily Usage Hrs	No. of Appliances																				
CFL	12	6	3																				
Fan	50	8	2																				
TV (21")	150	2	1																				
Computer	250	3	1																				
	b	With circuit diagram ,Explain the working of a Miniature Circuit Breaker	8																				
	C	Write a short note on types of wiring.	4																				
OR																							
5	a	With circuit diagram and functional table, Explain the control of lamp using 2 and 3 switches.	6																				
	b	Why pipe earthing is preferred more than plate earthing for more protection. With diagram explain pipe earthing.	6																				
	c	An electrical installation consists of 12 light points of 80 W each, 10 light points of 50 W lamp, 2 fans of 100 W capacity and a pump motor of 1.5 HP. Assuming that 40% of light and fans are used for 5 hours per day and that the water pump works for 2 hours	6																				

		daily. Find out the monthly consumption and cost of electricity bill, based on the tariff of Rs. 4.5 per Kwh. (1HP=736 Watts).	
		UNIT-IV	
6	a	Assuming $\beta=200$, plot the Q point (I_C and V_{CE}) on the load line for the circuit of Figure 7.a. Also determine the value of V_B .  Figure 7.a	10
	b	Explain the different types of temperature sensors.	8
		OR	
7	a	With circuit diagram, Explain the operation of a RC coupled amplifier.	10
	b	Analyze the transistor biasing using voltage divider circuit.	8
		UNIT-V	
8	a	Explain the concept of power generation using RES solar and wind through block diagrams.	10
	b	What are the advantages (Any 4) and disadvantages (Any 4) of EV's?	8
		OR	
9	a	With diagram, Explain the function of various components of electric vehicle.	10
	b	Explain the National Policy for adoption of EVs	8

Signature of Scrutinizer:

Signature of Chairman

Name:

Name:

Course Code:	EE112AT
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PART-A

Q.No	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
B T	3	1	3	3	2	1	1			
COs	1	1	2	2	3	4	3			

PART-B

Question No		B T Levels	Cos addressed		Question No		BT Levels	Cos addressed
2	a	3	1		3	a	3	2
	b	3	1			b	3	2
	c	2	1			c	3	2
	d					d		
4	a	3	1		5	a	3	2
	b	4	2			b	2	3
	c	3	3			c	1	2
	d					d		
6	a	2	3		7	a	3	3
	b	2	3			b	2	3
	c					c		
	d					d		
8	a	2	3		9	a	2	4
	b	3	3			b	2	4
	c					c		
	d					d		
10	a	2	4					
	b	2	4					
	c							
	d							

Signature of Scrutinizer:

Signature of Chairperson:

Name:

Name:

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RV COLLEGE OF ENGINEERING®

Autonomous Institution affiliated to VT

I Semester B.E. January-2026 Examinations

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

BASICS OF ELECTRICAL ENGINEERING

MODEL QUESTION PAPER**(2025 SCHEME)****Time: 03 Hours****Maximum Marks: 100****Instructions to candidates:**

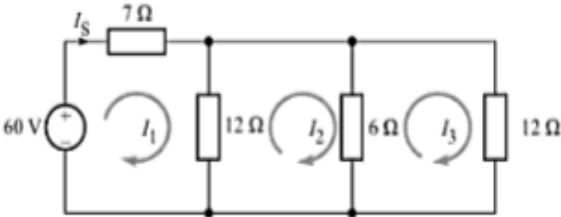
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 and 3 is compulsory. Answer any one full question from 4 or 5, 6 or 7 and 8 or 9.

PART-A

1	1.1	Mention the type of alternator used in nuclear power station and that in hydel power station.	1
	1.2	Mention any two advantages of three phase over single-phase AC voltage.	1
	1.3	Draw the phasor diagram and waveform of voltage and current of a RL series circuit.	1
	1.4	Mention the methods used to improve the power factor of the circuits.	1
	1.5	A 250 KVA, 11 KV/415V, 50 Hz, 1ϕ transformer has 80 turns on the secondary. Calculate rated primary and secondary currents	2
	1.6	The geyser is rated at 4kW, 230V, 50Hz. If it is switched ON for 50 minutes daily, what would be the energy cost saving per month, at the rate of Rs.7 per unit if it is replaced by a solar water heater?	2
	1.7	Given $R=20\Omega$, $L = 0.2H$ and $C = 100\mu F$ are connected in series across 220 V, 50 Hz supply. Calculate the current.	2

PART-B**UNIT-I**

2.a	With the help of a neat block diagram, explain the working principle of a Nuclear Power Plant. Also, mention its advantages and disadvantages.	6

b.	Apply mesh analysis to determine current drawn from the source in the network shown in Fig. 2.b 	8
c	State and Explain KCL and KVL	4

UNIT-II		
3a	A pure inductive coil allows a current of 10 A to flow from a 230V, 50 Hz supply. Find i) Inductance of the coil ii) Power absorbed and iii) equations for voltage and current.	8
b	Derive the expression of power in a RLC series circuit. Also draw the relevant phasor diagram and waveforms.	6
3.c	Mention the advantages of 3 ϕ System over single-phase system.	4
UNIT-III		
4.a	Derive the Torque equation of DC Motor and explain different types of torques in DC Motors	8
b	With necessary equations draw the three important characteristics of DC Shunt motor and mention the applications of the same.	6
c	A 4 pole generator with wave wound armature has 51 slots, each having 24 conductors. The flux per pole is 0.01 Weber. At what speed must the armature rotate to give an induced emf of 220 V? What will be the voltage developed if the winding is lap and the armature rotates at the same speed?	4
OR		
5.a	Draw the cross-sectional view of a D.C machine and explain the function of various parts of the same.	10
b	A 6 pole, Lap connected DC Shunt motor takes 300A at a speed of 400 RPM. The flux per pole is 75 mWb. The number of armature conductors is 500. The torque lost in friction, windage and iron losses can be assumed at 2.5%. Calculate i) The torque developed by the armature ii) Shaft Torque iii) Shaft Power in kW.	08
UNIT-IV		
6.a	transformer is rated at 100KVA, at full load $P_{cu}=1200W$ and $P_i=960W$. Calculate a) Efficiency at full load , U.P.F b) Efficiency at half load, 0.8 P.F. c) The load KVA at which η_{max} occurs.	8
b	Distinguish between salient and non salient alternators.	6

c	Derive emf equation of the 1ϕ transformer	4																				
OR																						
7.a	A 50KVA single phase transformer has an efficiency of 98% at full-load 0.8 p.f and 97% at 1/2 full load 0.8 p.f. Determine the full copper loss and iron loss.	8																				
b	With neat sketches, explain different types of rotors used in Induction motors	6																				
c	With neat sketch explain the constructional details of core type and shell type transformers.	4																				
UNIT-V																						
8.a	Estimate the electricity bill for the following loads <table><thead><tr><th>Name of the Appliance</th><th>Power Rating (W)</th><th>Avg. Daily Usage Hrs</th><th>No. of Appliances</th></tr></thead><tbody><tr><td>CFL</td><td>12</td><td>6</td><td>3</td></tr><tr><td>Fan</td><td>50</td><td>8</td><td>2</td></tr><tr><td>TV (21")</td><td>150</td><td>2</td><td>1</td></tr><tr><td>Computer</td><td>250</td><td>3</td><td>1</td></tr></tbody></table> Take electricity cost to be Rs.6 per unit.	Name of the Appliance	Power Rating (W)	Avg. Daily Usage Hrs	No. of Appliances	CFL	12	6	3	Fan	50	8	2	TV (21")	150	2	1	Computer	250	3	1	08
Name of the Appliance	Power Rating (W)	Avg. Daily Usage Hrs	No. of Appliances																			
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TV (21")	150	2	1																			
Computer	250	3	1																			
b	What is the necessity of earthing the electrical appliances? Explain with diagram plate earthing.	10																				
OR																						
9.a	Explain the safety precautions against electric shock	4																				
b	Distinguish between fuse and MCB	4																				
c	Explain with neat circuit diagrams the working of two-way and three-way control of a load, highlighting the differences between them and mentioning their practical applications.	10																				

Course Code: EE113ATD Course Title: Basics of Electrical Engineering

PART-A

Q.No	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
B T	1	2	3	1	2	2	2			
COs	1	1	2	3	2	4	3	3	3	4
Q No	1.11	1.12	1.13	1.14	1.15	1.16	1.17	1.18	1.19	1.20
B T										
COs										

PART-B

Question No		B T Levels	Cos addressed		Question No		BT Levels	Cos addressed
2	a	2	1		3	a	3	2
	b	3	2			b	2	2
	c	2	1			c	1	2
	d					d		
4	a	3	3		5	a	2	3
	b	2	3			b	3	3
	c					c		
	d					d		
6	a	2	3		7	a	2	3
	b	2	3			b	2	3
	c	1	3			c	1	2
	d					d		
8	a	3	4		9	a	2	4
	b	2	4			b	1	4
	c					c	2	4
	d					d		

Signature of Scrutinizer:
Name:

Signature of Chairman
Name:

USN

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RV COLLEGE OF ENGINEERING®
Autonomous Institution affiliated to VTU
First Semester B. E.

DEPARTMENT OF MATHEMATICS
Fundamentals of Linear Algebra, Calculus and Numerical Methods
MODEL QUESTION PAPER - I
Branches: ECE, ETE, EEE

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 and 3 are compulsory. Answer any one full question from 4&5, 6&7, 8 and 9.

PART-A

1	1.1	Given $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & -1 & 5 \\ 3 & 0 & 3 \end{bmatrix}$, then the eigen values of the matrix A^{-2} are _____.	1
	1.2	If rank of the matrix $A = \begin{bmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ k & 13 & 10 \end{bmatrix}$ is 2, then the value of $k =$ _____.	1
	1.5	The Maclaurin series expansion for $e^{-\frac{x}{2}}$ is _____.	2
	1.6	If $w(x, y) = x^2y$ then $\frac{\partial^2 w}{\partial x \partial y}$ at the point $(-1, 1) =$ _____.	2
	1.7	Evaluate $\int_0^1 \int_2^3 xy \, dy \, dx$.	2
	1.9	Given $y(2) = -2, y(4) = 4, y(6) = 10, y(8) = 12, y(10) = 14$, then $\nabla^3 y(8) =$ _____.	2

PART-B

2	a	Investigate for what values of λ and μ the system of linear equations $2x + 3y + 5z = 9, 7x + 3y - 2z = 8, 2x + 3y + \lambda z = \mu$ has (i) no solution (ii) a unique solution (iii) an infinite number of solutions.	6
	b	The currents I_1, I_2, I_3 in an electrical network follow the linear equations $8I_1 - 3I_2 + 2I_3 = 20, 4I_1 + 11I_2 - I_3 = 33, 2I_1 + I_2 + 4I_3 = 12$. Compute I_1, I_2, I_3 using Gauss-Seidel iteration method. Perform five iterations.	6
	c	Eigen values and eigen vectors are used to calculate the theoretical limit of how much information can be carried via a communication channel. Identify the dominant eigen value and corresponding eigen vectors of the matrix $A = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix}$ by taking the initial approximation $\begin{bmatrix} 1 \\ 0.8 \\ 0.8 \end{bmatrix}$. Perform 5 iterations.	6
3	a	Find the slope of the tangent to the parabola $\frac{2a}{r} = 1 - \cos \theta$ at $\theta = \frac{2\pi}{3}$.	6
	b	Show that the radius of curvature at any point (r, θ) on the Cardioid $r = a(1 - \cos \theta)$ varies as $\sqrt{r}$.	6

	c	Obtain the circle for curvature of the curve $\sqrt{x} + \sqrt{y} = 4$ at the point where it cuts the line passing through the origin making an angle 45° with x-axis.	6
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4	a	Prove that $v = e^{a\theta} \cos(a \log r)$ is the solution of the Laplace equation $v_{rr} + \frac{1}{r} v_r + \frac{1}{r^2} v_{\theta\theta} = 0.$	8
	b	In robotics, the functions representing robotic arm from cartesian to any system (x, y) are given by $x = e^u \cos v$ and $y = e^u \sin v$. As the Jacobian represents transformation factor between different systems, verify $\frac{\partial(x,y)}{\partial(u,v)} \times \frac{\partial(u,v)}{\partial(x,y)} = 1$.	10

OR			
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5	a	(i) If $z = f(y - 3x) + g(y + 2x) + \sin x - y \cos x$, show that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6 \frac{\partial^2 z}{\partial y^2} = y \cos x.$ (ii) Find $\frac{du}{dt}$ given $u = \sin\left(\frac{x}{y}\right)$ where $x = e^t, y = t^2$.	8
	b	The temperature T at any point (x, y, z) in space is $T = 600xyz^2$. Find the highest temperature on the unit sphere using Lagrange multiplier.	10

OR			
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6	a	Change the order of integration and evaluate $\int_1^0 \int_x^{\sqrt{2-x^2}} \frac{x}{\sqrt{x^2+y^2}} dy dx$.	8
	b	Determine the centre of gravity of the triangular lamina bounded by the coordinate axis and the line $x + y = 1$.	10

OR			
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7	a	Show that $\int_0^{\frac{\pi}{2}} \int_0^{a \sin \theta} \int_0^{\frac{a^2-r^2}{a}} r dz dr d\theta = \frac{5\pi a^3}{64}$.	8
	b	Find the volume of the tetrahedron $x \geq 0, y \geq 0, z \geq 0, \frac{x}{a} + \frac{y}{b} + \frac{z}{c} \leq 1$.	10

8	a	In an experiment, the values of the output y were recorded for input x from 1.0 to 3.5 at intervals of 0.5. Estimate by using appropriate interpolation formulae, the values of y for input (i) $x = 1.2$, (ii) $x = 3.4$ <table><tr><td>x</td><td>1.0</td><td>1.5</td><td>2.0</td><td>2.5</td><td>3.0</td><td>3.5</td></tr><tr><td>y</td><td>177</td><td>166</td><td>146</td><td>130</td><td>115</td><td>102</td></tr></table>	x	1.0	1.5	2.0	2.5	3.0	3.5	y	177	166	146	130	115	102	8
	x	1.0	1.5	2.0	2.5	3.0	3.5										
y	177	166	146	130	115	102											
b	The following data defines the sea-level concentration of dissolved oxygen for fresh water as a function of temperature: <table><tr><td>T (°C)</td><td>0</td><td>8</td><td>15</td><td>25</td><td>32</td></tr><tr><td>O₂ (mg/L)</td><td>14.621</td><td>11.843</td><td>9.870</td><td>8.418</td><td>7.305</td></tr></table> Calculate the amount of oxygen when temperature is 10°C and 35°C.	T (°C)	0	8	15	25	32	O ₂ (mg/L)	14.621	11.843	9.870	8.418	7.305	10			
T (°C)	0	8	15	25	32												
O ₂ (mg/L)	14.621	11.843	9.870	8.418	7.305												

OR			
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9	a	Numerical integration is used to simulate devices such as semiconductors and	8
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		antennas. Estimate the value of the integral $\int_0^2 e^{-x^2} dx$ using Simpson's 1/3, Simpson's 3/8 and Weddle's rules, by dividing the interval [0,2] into six equal sub intervals.	
	b	Numerical integration is used to simulate devices such as semiconductors and antennas. Estimate the value of the integral $\int_0^2 e^{-x^2} dx$ using Simpson's 1/3, Simpson's 3/8 and Weddle's rules, by dividing the interval [0,2] into six equal sub intervals.	10

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

I Semester B. E.(AS, CH, IEM,ME) (2025 SCHEME)

Department of Mathematics

FUNDAMENTALS OF LINEAR ALGEBRA, CALCULUS AND DIFFERENTIAL EQUATIONS

MODEL QUESTION PAPER**Time: 03 Hours****Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 and 3 are compulsory. Answer any one full question from 4 or 5, 6 or 7 and 8 or 9.
3. Use of first year Mathematics Handbook for B.E. is permitted. Do not write anything on handbook.

Part - A			
1	1.1	Rank of the matrix $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 5 & 8 \\ -3 & 4 & 4 \end{bmatrix}$ is ____.	01
	1.2	The eigenvalues of a matrix B are 4, 1 and 0.5 then the determinant of the matrix B are ____.	01
	1.3	A particle moves along a circle $r = a \cos \theta$, the angle between radius vector and tangent to the curve at the point (r, θ) is ____.	02
	1.4	If $w(x, y) = \sin(xy)$, then $\frac{\partial^2 w}{\partial x \partial y}$ at the point $(\pi, 1) =$ ____.	02
	1.5	Sketch the domain of integral $\int_0^1 \int_{x^2}^x f(x, y) dy dx$.	02
	1.6	The differential equation whose auxiliary equation has roots -1, 2, 2 is ____.	02

Part - B

Unit-I			
2	a	Investigate for what values of λ and μ the system of linear equations $2x + 3y + 5z = 9$, $7x + 3y - 2z = 8$, $2x + 3y + \lambda z = \mu$ has (i) no solution (ii) a unique solution (iii) an infinite number of solutions.	06
	b	The temperatures x, y, z of a metal plate follows the linear equations $5x + 2y + z = 12$; $x + 4y + 2z = 15$; $x + 2y + 5z = 20$. Compute x, y, z using by Gauss Seidel iteration method. Perform 5 iterations by taking initial approximation to the solution as $(1, 0, 3)$.	06
	c	Determine the numerically largest Eigen value and the corresponding Eigen vector of the matrix $\begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix}$ by taking the initial approximation to the Eigen vector as $(1, 0, 0)^T$.	06

Unit-II			
3	a	Determine the angle of intersection of the curves $r = \sin \theta + \cos \theta$ and $r = 2 \sin \theta$.	06

	b	Show that the radius of curvature at (3,7) on $x = 3 + 4 \cos t$, $y = 4 + 3 \sin t$ is $\frac{16}{3}$.	06
	c	Expand $\log(1 + e^x)$ in ascending powers of x as far as the term containing x^4 .	06

Unit-III			
4	a	The steady state temperature of a circular metal plate is $u = r^2 \cos(2\theta) - r \cos(\theta) + 2$. Show that u satisfies the Laplace equation $u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} = 0$.	08
	b	Compute $J\left(\frac{x,y,z}{u,v,w}\right)$ and $J\left(\frac{u,v,w}{x,y,z}\right)$ given $x = u$, $y = u \tan v$, $z = w$. Hence verify $JJ' = 1$.	10
OR			
5	a	If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$, then prove that $6 \frac{\partial u}{\partial x} + 4 \frac{\partial u}{\partial y} + 3 \frac{\partial u}{\partial z} = 0$.	08
	b	Show that by Lagrange's method of undetermined multipliers, the rectangular solid of maximum volume that can be inscribed in a sphere is a cube.	10

Unit-IV			
6	a	Find the area bounded by one quadrant of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	08
	b	Evaluate $\int_0^1 \int_{e^x}^e \frac{dx dy}{\log y}$ by changing the order of integration.	10
OR			
7	a	Find the value of $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x + y + z) dy dx dz$.	08
	b	Calculate the center of gravity by double integration of a lamina $x = a \cos^3 t$, $y = a \sin^3 t$ in first quadrant, the density being $\rho = kxy$, where k is a constant.	10

Unit-V			
8	a	Solve $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = 8(x^2 + e^{2x} + \sin 2x)$.	08
	b	Apply the method of variation of parameters to solve the ordinary differential equation $\frac{d^2 y}{dx^2} + \frac{dy}{dx} = \operatorname{cosec} x$	10
OR			
9	a	An electric circuit consists of an inductance of 0.1 Henry, a resistance of 20 Ohms and a condenser of capacitance 25 micro-Farads connected in series. Find the charge q and the current i at any time t . Given that $L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = 0$ at $t = 0$, $q = 0.05$ C, $i = \frac{dq}{dt} = 0$ when $t = 0$.	08
	b	Solve $x^3 \frac{d^3 y}{dx^3} + 3x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + 8y = \cos(\log x)$.	10

RV COLLEGE OF ENGINEERING®
(An Autonomous Institution Affiliated to VTU)
I Semester B. E. January-2026 Examinations
Department of Mathematics
APPLIED MATHEMATICS - I
(2025 SCHEME)

Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 and 3 are compulsory. Answer any one full question from 4 or 5, 6 or 7 and 8 or 9.
3. Marks will not be awarded for direct answers without supporting steps.
4. Write answers up to four decimal places.
5. Use of first year Mathematics Handbook for B.E. is permitted. Do not write anything on handbook.

Part - A			
1	1.1	If A is a 3×4 matrix then rank of A cannot exceed _____.	01
	1.2	If 2 and -5 are the eigenvalues of a matrix A , then the eigenvalues of A^4 are _____.	01
	1.3	If $x^3 + y^3 - 3axy = 0$ find $\frac{dy}{dx}$.	02
	1.4	Evaluate $\int_1^4 \int_3^5 x^2 y \, dy \, dx$.	02
	1.5	Solve: $\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 7y = 0$.	02
	1.6	Determine mean of x and y if two regression equations of the variables x and y are $8x = 10y - 66$, $18y = 40x - 214$.	02

Part - B

Unit-I			
2	a	Investigate the values of a and b for which the system of equations $x + 2y + 3z = 6$, $x + 3y + 5z = 9$, $2x + 5y + az = b$, has a (i) Unique solution (ii) Infinite solution (iii) No solution. Find the solution in the case (i) and (ii).	08
	b	Use Gauss-Seidel method to compute x , y and z for the system of equations: $8x - 3y + 2z = 20$, $6x + 3y + 12z = 35$, $4x + 11y - z = 33$. Carry out four iterations.	05
	c	Obtain the numerically largest eigenvalue and corresponding eigenvector by power method for the matrix $A = \begin{bmatrix} 10 & 2 & 1 \\ 2 & 10 & 1 \\ 2 & 1 & 10 \end{bmatrix}$. Choose the initial vector as $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$. Perform 3 iterations.	05

Unit-II			
3	a	If $\theta = t^n e^{-\frac{r^2}{4t}}$ find the value of n such that $\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = \frac{\partial \theta}{\partial t}$.	08
	b	Jacobian represents transformation function to traverse between coordinate system. If $x = u(1-v), y = uv$ then compute $\frac{\partial(x,y)}{\partial(u,v)}$.	05
	c	Divide 24 into three parts such that the continued product of I, square of II and cube of III is maximum.	05

Unit-III			
4	a	Evaluate $\int_0^1 \int_x^{\sqrt{x}} xy \, dy \, dx$.	06
	b	Change the order of integration in $\int_0^a \int_0^{2\sqrt{ax}} x^2 \, dy \, dx$ and hence evaluate.	06
	c	Evaluate $\int_0^1 \int_0^x \int_0^{x+y} (x+y+z) \, dz \, dy \, dx$.	06
OR			
5	a	Find the area between the parabola $y = 4x - x^2$ and the line $y = x$, using double integration.	05
	b	Evaluate $\int_{-a}^a \int_0^{\sqrt{a^2-x^2}} \sqrt{x^2+y^2} \, dy \, dx$ by changing to polar coordinates.	05
	c	Find the center of gravity $(\bar{x}, \bar{y})$ of a mass of density $f(x,y) = 1$ in the region R the semidisk $x^2 + y^2 \leq a^2, y \geq 0$.	08

Unit-IV			
6	a	Solve the initial value problem $2x''(t) + 5x'(t) - 3x(t) = 0$, given that $x(0) = 1, x'(0) = 4$.	05
	b	Solve $(D-2)^2 y = 8(e^{2x} + \sin 2x)$.	05
	c	Solve $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + y = \log(x)$.	08
OR			
7	a	Solve $y'' - 2y' + y = xe^x \sin x$.	05
	b	Solve: $y'' + a^2 y = \sec(ax)$ by the method of variation of parameters.	05
	c	Find the charge on the capacitor $q = q(t)$, in a LRC series circuit from $L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E(t)$, when $L = (5/3)\text{h}$, $R = 10\Omega$, $C = (1/30)\text{f}$, $E(t) = 300\text{V}$, $q(0) = 0$, $\frac{dq}{dt} = 0$.	08

Unit-V																						
8	a	From the following frequency distribution compute first four moments about the mean and also find the measures β_1 and β_2 .					09															
		<table><tr><td>x</td><td>4</td><td>8</td><td>12</td><td>16</td><td>20</td></tr><tr><td>f</td><td>2</td><td>2</td><td>1</td><td>4</td><td>1</td></tr></table>						x	4	8	12	16	20	f	2	2	1	4	1			
x	4	8	12	16	20																	
f	2	2	1	4	1																	
	b	Find by the method of least squares the best values of a, b, c for the equation $P = a + bW + cW^2$ which fits the data					09															
		<table><tr><td>W</td><td>20</td><td>40</td><td>60</td><td>80</td><td>100</td><td>120</td></tr><tr><td>P</td><td>5.5</td><td>9.1</td><td>14.9</td><td>22.8</td><td>33.3</td><td>46</td></tr></table>						W	20	40	60	80	100	120	P	5.5	9.1	14.9	22.8	33.3	46	
W	20	40	60	80	100	120																
P	5.5	9.1	14.9	22.8	33.3	46																
OR																						

9	a	Find the coefficient of correlation between industrial production and export using the following data and comment on the result.	09																		
		<table><tr><td>Production in crore</td><td>55</td><td>56</td><td>58</td><td>59</td><td>60</td><td>61</td><td>62</td></tr><tr><td>Export in crore tons (y)</td><td>35</td><td>38</td><td>38</td><td>39</td><td>44</td><td>43</td><td>45</td></tr></table>		Production in crore	55	56	58	59	60	61	62	Export in crore tons (y)	35	38	38	39	44	43	45		
Production in crore	55	56		58	59	60	61	62													
Export in crore tons (y)	35	38	38	39	44	43	45														
	b	The latent heat of vaporization of steam r is given in the following table at different temperatures t :	09																		
		<table><tr><td>t</td><td>40</td><td>50</td><td>60</td><td>70</td><td>80</td><td>90</td><td>100</td><td>110</td></tr><tr><td>r</td><td>1069.1</td><td>1063.6</td><td>1058.2</td><td>1052.7</td><td>1049.3</td><td>1041.8</td><td>1036.3</td><td>1030.8</td></tr></table>		t	40	50	60	70	80	90	100	110	r	1069.1	1063.6	1058.2	1052.7	1049.3	1041.8	1036.3	1030.8
t	40	50		60	70	80	90	100	110												
r	1069.1	1063.6	1058.2	1052.7	1049.3	1041.8	1036.3	1030.8													
		Fit a relation of the form $r = a + bt$.																			

Signature of Scrutinizer:

Signature of Chairman:

Name:

Name:

USN

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RV COLLEGE OF ENGINEERING
Autonomous Institution affiliated to VTU
I/II Semester B.E. January -2026 Examinations
DEPARTMENT OF MECHANICAL ENGINEERING
COURSE TITLE: ELEMENTS OF MECHANICAL ENGINEERING

MODEL QUESTION PAPER

(2025 SCHEME)

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 and 3 are compulsory. Answer any one full question from 4 and 5, 6 and 7, and 8 and 9.
3. Use of Steam Tables is permitted

PART-A

			M	L	CO
1	1.1	For dry saturated steam, the dryness fraction is equal _____	01	1	1
	1.2	The main function of nozzle in impulse turbine is _____	01	1	2
	1.3	In an IC Engine, Piston rings are provided to maintain _____	01	1	3
	1.4	Stroke of the piston will be equal to _____ the radius of the crank.	01	1	3
	1.5	In hybrid electric vehicles, _____ converts AC or DC electrical energy into AC energy suitable for the operation of the electric motor.	01	1	3
	1.6	Polyester is the example of _____ type of polymer	01	1	2
	1.7	_____ type of control systems doesn't have feedback system.	01	1	2
	1.8	_____ is the type of filler material commonly used in soldering	01	1	2
	1.9	The purpose of ABS in vehicles is to _____	01	1	2
	1.10	_____ is an example of thermoplastics	01	1	1

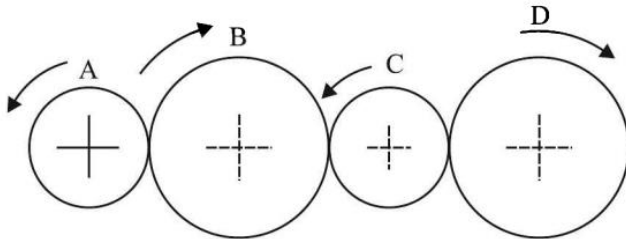
PART-B

UNIT-I

2	a	Define Engineering materials. Give the detailed classification of materials along with their applications	08	2	2
	b	Discuss in details physical, mechanical & electrical properties of the materials.	10	3	2

UNIT-II

3	a	Briefly explain Sand Casting with a neat sketch showing all the important parts of the sand casting mold	06	2	1
	b	How do you carry out the Electric arc welding process? Explain with neat process sketch and the safety measures while executing.	06	3	2
	c	With the detail diagram, explain the various elements of CNC Machine.	06	2	3

UNIT-III					
4	a	Determine the enthalpy, volume, internal energy, and entropy of superheated steam at 14 bar pressure and 200°C. Neglect the volume of water and take specific heat of superheated steam equal to 2.1 kJ/kg.	08	3	4
	b	Differentiate between Impulse steam turbine and Reaction steam turbine with the help of velocity and pressure diagrams across the nozzle and rotor.	10	3	4
OR					
5	a	Identify the hydraulic turbine which resembles the propeller of a Ship. Explain its working principle with a neat labelled diagram	08	3	4
	b	Identify the refrigeration system that works on II law of thermodynamics and which uses Freon 12 as a working medium to circulate between all the parts of the system and explain its working principle using a neat diagram marking all important components.	10	3	4
UNIT-IV					
6	a	Explain with schematic sketches working principle of IC Engine in which burning of fuel take place at constant volume and crank shaft rotates two revolutions for every cycle.	08	3	3
	b	A four-cylinder four-stroke diesel engine has bore of 100 mm, stroke of 120 mm and piston speed of 10 m/s. The engine develops BP of 20 kW per litre of cylinder stroke volume. Brake thermal efficiency of the engine is 30% with a fuel of CV of 40 MJ/kg and specific gravity of 0.90. Determine (i) rpm, (ii) BP, and (iii) engine fuel requirements, in litres/h	10	3	3
OR					
7	a	Explain Simple and compound gear train with a neat sketch and derive for the velocity ratio for the above-mentioned gear trains	10	2	2
	b	A simple gear train as shown in Figure. has four gears namely A, B, C and D. Gear A is driver gear and D gear is driven gear. Gears B and C are intermediate gears. Driving gear A rotates anticlockwise direction with a speed of 300 rpm. The number of teeth on gears A, B, C and D are 30, 90, 60 and 120, respectively. Determine (i) the speed of the driven gear and the sense of rotation, (ii) the speed and direction of rotations of intermediate gears, (iii) the velocity ratio of gear trains and (iv) the train value 	08	3	3
UNIT-V					
8	a	Elucidate the working of a series Hybrid vehicle with a neat sketch	08	2	2
	b	Describe the basic structure and anatomy of industrial robots. Explain how these contribute to the movement and functionality	10	2	3
OR					
9	a	Classify and explain the types of robots based on configurations.	08	2	4
	b	Enumerate mechatronics control system using washing machine as an example with an appropriate diagram	10	2	4



RV College of Engineering®

Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Course Code: PY2111A

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RV COLLEGE OF ENGINEERING
Autonomous Institution affiliated to VTU

I Semester B.E. January-2026 Examinations

DEPARTMENT: PHYSICS

COURSE TITLE: CONDENSED MATTER PHYSICS FOR ENGINEERS

Time: 3 hours

Maximum marks: 100

Instructions to Candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 & 3 is compulsory. Answer any one full question from 4 or 5, 6 or 7 and 8 or 9.

PART-A

1		Questions	M	BT	CO
	1.1	What are Fermions?	1	1	1
	1.2	Find the sensitivity of a temperature sensor whose output is 700 mV at 20 °C and 900 mV at 60 °C.	2	3	3
	1.3	Draw the energy level diagram of an n type semiconductor.	1	1	2
	1.4	Two particles confined in adjacent quantum wells can interfere with each other to form bands. This is possible due to which two principles of Quantum Physics?	2	3	4
	1.5	Write two properties of an Eigen function.	1	1	1
	1.6	Name the emission responsible for lasing action.	1	1	1
	1.7	In Andreev reflection, what is reflected from the interface in the normal metal side and what is transmitted in the superconducting region?	2	1	1

PART-B

Unit I			M	BT	CO
2	a	Sketch and explain the variation of Fermi factor in metals for energy levels below and above Fermi energy at zero Kelvin. Explain with a neat sketch, the variation of Fermi level with temperature in a p-type semiconductor.	8	3	2
	b	Show that the Fermi level in an intrinsic semiconductor lies at the center of band gap. Sketch and explain the effect of dopant concentration on the Fermi level in an n-type semiconductor.	8	2	1
	c	In a Hall effect experiment, a current of 0.35 A is passed through a semiconductor slab of width 5 mm and thickness 1 mm. The Hall voltage measured is 0.2mV in the presence of magnetic field of 0.5 T. Calculate Hall coefficient, carrier concentration and drift velocity.	2	3	3



Unit II			M	BT	CO
3	a	Explain how a p-n junction diode can be used as a bridge rectifier. Plot the input and output waveforms for the same. Sketch the I-V characteristics of Zener diode.	8	3	1
	b	What is a thermistor? Write a short note on types of thermistors based on their temperature coefficients. Sketch the energy band diagrams of p-n junction under no bias, forward bias and reverse bias condition.	8	2	2
	c	Pure silicon at 300 K has equal electron and hole concentration of $1.5 \times 10^{16} m^{-3}$. Doping by indium increases hole concentration to $4.5 \times 10^{22} m^{-3}$. Find the electron concentration in the doped silicon.	2	3	3

Unit III			M	BT	CO
4	a	Derive the time independent Schrodinger wave equation starting from the classical wave equation.	6	1	2
	b	Use the one-dimensional finite well located symmetrically around the origin to arrive at the transcendental equation for only the even symmetry solution of the problem.	8	2	2
	c	Calculate the difference between the ground state and 1 st excited state of an infinite well in one, two and three dimensions.	4	3	3
OR					
5	a	Apply the time independent Schrodinger equation to arrive at an expression for the eigenvalues of a particle trapped in a one dimensional infinite potential well.	8	2	2
	b	State Heisenberg's Uncertainty Principle. With a neat labelled diagram, arrive at an expression for broadening of spectral lines for a two level atomic system. How is this principle applicable for LASER.	8	1	2
	c	The de-Broglie wavelength associated with an electron is 1nm. Find the potential difference by which the electron is accelerated.	2	1	3

Unit IV			M	BT	CO
6	a	With a clear diagram, explain how light rays propagate through an optical fiber. Define Acceptance angle of an optical fibre. Obtain an expression for acceptance angle of an optical fiber in terms of refractive indices of the fibre.	8	2	2
	b	Explain the construction and working of semi-conductor diode laser with suitable diagrams.	8	2	2
	c	If a laser cavity has a length of 0.1 m and the number of modes excited is 10^6 , find the operating wavelength of the cavity.	2	3	3
OR					
7	a	Elaborate on the construction and propagation mechanism of signal in a graded-index multimode fiber using suitable diagrams. Write three main differences between single mode and multimode fibers.	8	2	2
	b	Arrive at an expression for energy density in-terms of Einstein's co-efficient in a LASER system. Write a short note of an absorption-based optical attenuator.	8	3	1, 3
	c	An optical fiber has a core refractive index 1.48 and a cladding refractive index 1.46. The fiber is placed in a medium with refractive index 1.33. Calculate the acceptance angle of the fibre.	2	3	3



Unit V			M	BT	CO
8	a	Explain BCS theory of Superconductivity. What is Meissner effect? Write one applications of Meissner effect.	8	2	2
	b	With a labelled diagram, explain the construction and working of a dc squid.	6	2	2
	c	How do superconductors improve power transmission and medical imaging?	4	4	2
OR					
9	a	Explain Type I and Type II superconductors with relevant diagrams.	8	2	1
	b	With a labelled diagram, explain the construction and working of an ac squid.	6	2	2
	c	A type I superconductor has a critical magnetic field of 0.1 T at 0 K. What is the value of the magnetic field above which the material behaves as a normal metal at 4.2 K. The critical temperature of the material is 7.2 K.	4	3	3

RV COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU
I Semester B.E. January-2026 Examinations

DEPARTMENT OF PHYSICS

Model Question Paper

COURSE TITLE: Classical Physics for Engineers
(2025 SCHEME)

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

- 1) Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
- 2) Answer FIVE full questions from Part B. In Part B question numbers 2 & 3 are compulsory. Answer any one full question from 4 or 5, 6 or 7 and 8 or 9.

PART A

1.1	Mention the condition for the under damped oscillations of the spring mass system.	01
1.2	What is the effect of damping factor on the amplitude of forced oscillations.	01
1.3	How does the time period of a torsional pendulum depend on the rigidity modulus of the wire?	01
1.4	What happens to a material if it stretched beyond its elastic limit?	01
1.5	How joules heating is different from thermoelectric effect.	01
1.6	What is the ground state energy of an electron in an infinite potential well?	02
1.7	What are the benefits nondestructive testing of products?	01
1.8	Why [surface area/volume] ratio is very large for nanoparticles compared to bulk materials?	02

PART B

	Unit 1	
2a	Set up the differential equation for the damped oscillations and obtain an expression for displacement of the oscillations.	08
2b	Consider three springs of stiffness constant K_1 , K_2 and K_3 . Obtain the expression for effective spring constant with these springs considered in 1) Series, 2) Parallel.	06
2c	A body of mass 500 g is attached to a spring and the system is driven by an external periodic force of amplitude 15 N, and frequency 0.796 Hz. The spring extends by a length of 88 mm under the given load. Calculate the amplitude of oscillation if the resistance coefficient of the medium is 5.05 kg/s.	04

	Unit 2	
3a	Find an expression for the bending moment in a horizontal beam clamped at one end and loaded at the free end.	08
3b	Derive an expression for the time period of a body when it executes angular simple harmonic oscillations.	06
3c	A fixed beam of length 50 cm is depressed by 15 mm at the loaded end. Calculate the depression at a distance 0.3 m from the fixed end for the same load at the free end.	04
	Unit 3	
4a	Define Seebeck effect and Peltier effect. Explain the construction and working of a thermocouple to measure the absolute temperature of the system. Mention its advantages.	08
4b	Explain the construction and working of a thermoelectric generator.	06
4c	The thermo e.m.f of a Cu-Fe thermocouple is $2160 \mu\text{V}$ when the cold junction is at 0°C and the hot junction at 250°C . Calculate the constants a and b if the neutral temperature is 330°C .	04
	OR	
5a	Explain the porous plug experiment with necessary theory.	08
5b	Along with the principle, describe the construction and working of the Linde's air liquefier.	06
5c	The e.m.f. in micro volts (e) of a thermocouple, one junction of which is at 0°C is given by $e = 1600 T - 4 T^2$, where $T^\circ\text{C}$ is the temperature of hot junction. Find (i) neutral temperature (ii) Peltier coefficient.	04
	Unit 4	
6a	Using the time independent Schrodinger's wave equation, find the solutions for a particle in an infinite potential well of width ' L '. Hence obtain normalized wave function.	08
6b	Using the solutions of the time independent Schrodinger's equation, explain the energy quantization of an electron by restricting its position inside the infinite potential well. Explain the quantum tunneling of an electron in a potential well.	06
6c	Assuming a particle moving in one dimensional infinite potential box of width $25\text{\AA}$. Calculate the probability of finding the particle within a small interval of $0.05 \text{\AA}$ at the center of the box when it is in its state of least energy.	04
	OR	
7a	With the help of neat sketch, explain SOL-GEL synthesis for producing nanomaterials.	08
7b	With the help of neat sketch, explain laser ablation process for synthesis of nanopowders.	06
7c	Find the lowest energy of an electron confined to move in a one-dimensional box of length $1\text{\AA}$. Express the result in electron volts.	04

	Unit 5	
8a	What is non-destructive testing? Explain with principle how flaw in a solid can be detected by non-destructive method using ultrasonics.	08
8b	Explain in detail how liquid penetrant method is used in NDT. What are its advantages and limitations?	06
8c	A steel block of height 90 mm is X-rayed using the displacement method. The distance from the X-ray source to the film is 480 mm. The X-ray tube is moved by 30 mm between exposures. The separation between the images of a defect on the film is found to be 12 mm. Calculate the depth of the defect from the bottom surface of the steel block.	04
	OR	
9a	Explain the construction and working of scanning electron microscope with suitable diagrams.	08
9b	Explain the construction and working of a Bragg's X-ray diffractometer.	06
9c	In an x-ray diffraction, peak width at half maxima is 0.6° and its corresponding Bragg angle is 24° . Calculate the crystallite size using Scherrer equation. Given: Wavelength of x-ray used is $1.54 \text{ \AA}$, Scherrer constant = 0.9.	04

Signature of the Scrutinizer:

Signature of the Chairman:

Name

Name



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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU, Belagavi)
I Semester B. E. Model Question paper
Course: APPLIED PHYSICS FOR ENGINEERS

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. Question number 2 and 3 are compulsory. Choose any one full question from 4 or 5, 6 or 7 and 8 or 9.

PART-A

Question No.	Questions	M	CO	BT
1.1	What is meant by the sharpness of resonance?	01	2	2
1.2	What type of damping is preferred in automobile shock absorbers?	01	2	2
1.3	Define Poisson's ratio.	01	1	1
1.4	What is the neutral surface in bending of beams?	01	2	2
1.5	Name a natural piezoelectric material.	01	1	1
1.6	What is the basic function of a shape memory alloy.	01	1	1
1.7	Write Sabine's formula for reverberation time.	01	3	3
1.8	What is the main function of a noise barrier?	01	3	3
1.9	State Bragg's law of diffraction.	01	1	1
2.0	Mention the type of waves used in ultrasonic testing.	01	1	1

PART – B

Question No.	Questions.	M	CO	BT
2a	Derive the equation of motion for a damped harmonic oscillator. Discuss the different types of damping using graphical approach.	8	2	2
2b	What are forced harmonic oscillations and derive the amplitude and phase equations for forced harmonic oscillator.	6	2	2
2c	A forced oscillator is acted upon by a periodic driving force of frequency 10 rad/s. The natural frequency is 9.8 rad/s, and damping coefficient 0.2s ⁻¹ . Find the amplitude at resonance.	4	3	3
3a	With a neat diagram, explain the stress–strain curve for a ductile material and mark all important points.	8	2	2
3b	Derive the expression for the bending moment of a beam and define the neutral axis.	6	2	2
3c	A brass wire 2 m long and 1.2 mm in diameter is twisted through 1 radian. Find the couple required if rigidity modulus is 4×10 ¹⁰ Nm ⁻² .	4	3	3
4a	Explain the shape memory effect in alloys and describe the phase transformation mechanism involved.	8	2	2
4b	Explain the working principles, construction, and applications of temperature sensors:(a) Resistance Temperature Detector (RTD) (b) Thermistor.	6	2	2
4c	A Resistance Temperature Detector (RTD) has resistance 100Ω at 0°C and temperature coefficient 0.00385/°. At a certain temperature, the resistance is measured as 119.25 Ω. Find the temperature of the RTD, and the change in resistance per degree Celsius between 0°C and this temperature.	4	3	3
	OR			
5a	Describe the working principle and construction of a piezoelectric sensor. How is the piezoelectric effect utilized in vibration sensing applications in civil or automotive systems?	8	2	2
5b	Explain in detail the engineering applications of smart materials in civil and structural health monitoring systems.	6	1	1
5c	A quartz crystal generates a charge of 0.75 μC under a force of 15 N. The crystal's capacitance is 100 pF. Find charge sensitivity and output voltage developed.	4	3	3
6a	Define noise barriers and explain how their design parameters influence their effectiveness in reducing highway noise.	8	2	2
6b	Explain Doppler effect in sound and derive the expression for apparent frequency.	6	1	1
6c	A conference room has a volume of 4000 m ³ and an average absorption area of 600 m ² . How much additional absorption area must be added to reduce the reverberation time from 1.07 s to 0.8 s?	4	3	3
	OR			
7a	Explain the principle of sound absorption and differentiate between porous absorbers and panel absorbers in terms of their working principle, materials, and applications.	8	1	1
7b	Explain the design and applications of highway noise barriers and materials used for noise reduction.	6	2	2
7c	A stationary radar emits sound of frequency 10 kHz toward a car moving at 30 ms ⁻¹ . The sound reflects back from the car. Find the frequency difference between emitted and received sound. Velocity of sound in air $v=340\text{ms}^{-1}$.	4	3	3

8a	Describe the pulse-echo ultrasonic inspection method and explain how it helps in identifying flaws in materials.	8	2	2
8b	With neat diagrams, explain the construction and working of X-ray diffractometer.	6	1	1
8c	The XRD peak of a Nickel oxide nanoparticle is observed at $2\theta=43.2^\circ$ with a full width at half maximum (FWHM) of 0.15° . Using Cu K α radiation ($\lambda=1.5406 \text{ \AA}$), estimate the crystallite size.	4	3	3
OR				
9a	With a neat sketch, describe the principle, construction, and working of Ultrasonic Testing. Explain pulse-echo method and its advantages.	8	2	2
9b	Describe in detail the principle, construction, and working of Differential Scanning Calorimeter with applications.	6	1	1
9c	In an X-ray diffraction experiment on a cubic crystal, the (200) diffraction peak is observed at an angle of $2\theta=64.5^\circ$ using Cu K α radiation of wavelength $1.54 \text{ \AA}$. Calculate the lattice constant of a crystal for first-order reflection.	4	3	3

Signature of the Scrutinizer

Signature of the Chairman

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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU, Belagavi)
I Semester B. E. Model Question paper
Course: Physics of Electrical and Electronic Materials

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. Question number 2 and 3 are compulsory. Choose any one full question from 4 or 5, 6 or 7 and 8 or 9.
3. **Physical constants:** Planck's constant = $6.625 \times 10^{-34} \text{Js}$,
Velocity of light = $3 \times 10^8 \text{m/s}$,
Boltzmann constant = $1.38 \times 10^{-23} \text{JK}^{-1}$,
Charge of electron = $1.6 \times 10^{-19} \text{C}$,
Mass of an electron = $9.1 \times 10^{-31} \text{kg}$,
Permittivity of free space = $8.854 \times 10^{-12} \text{Fm}^{-1}$

PART-A

Question No.	Questions	M	CO	BT
1.1	How would the hysteresis loop change, if a material transits from soft to hard magnetic characteristics due to alloying?	01	2	2
1.2	If the depth of a finite potential well is doubled, what qualitative changes would you expect in the energy levels?	01	2	2
1.3	State Matthiessen's rule of resistivity.	01	1	1
1.4	Why do Type II superconductors allow partial magnetic field penetration, whereas Type I does not?	01	2	2
1.5	Mention one application of thermoelectric generators.	01	1	1
1.6	Which type of impurities are added to produce n type semiconductors?	01	1	1
1.7	A ferromagnetic material has a magnetic field strength of 800 A/m . At 310 K , its magnetic susceptibility is 3.0×10^{-3} . Calculate magnetization and magnetic flux density.	02	3	3
1.8	Calculate the de Broglie wavelength for a beam of electrons whose energy is 45 eV ?	02	3	3

Question No.	Questions.	M	CO	BT
2a	With the help of domain theory, explain ferromagnetic hysteresis. Derive an expression for energy loss per cycle and explain how hysteresis loops help in selecting materials for transformers and permanent magnets.	8	2	2
2b	Derive the Clausius–Mossotti relation for elemental solid dielectric materials.	6	2	2
2c	The atomic weight and density of sulphur are 32 g/mol and 2.08 g/cm ³ respectively. If the electronic polarizability is $3.28 \times 10^{-40} \text{ F}\cdot\text{m}^2$, find the relative permittivity of sulphur.	4	3	3
3a	Derive the expressions for the energy eigenvalue and normalized wave function for a particle confined in a one-dimensional infinite potential well. Discuss how energy levels depend on the size of the well.	8	2	2
3b	Set up one dimensional time independent Schrodinger wave equation.	6	2	2
3c	An electron is trapped in a 1-dimensional infinite potential well of width $2 \times 10^{-10} \text{ m}$. Find the energy of the electron in the first and third quantum states.	4	3	3
4a	Define density of states. Using the same derive the expression for electron concentration in an intrinsic semiconductor.	8	2	2
4b	Explain the concept of Hall effect. With a neat sketch, derive expression for Hall voltage and Hall coefficient for n type semiconductor.	6	2	2
4c	A silicon sample shows a Hall coefficient $R_H = 3.66 \times 10^{-4} \text{ m}^3/\text{C}$ when a magnetic field of 0.5 T is applied. Find the carrier concentration and specify the type of semiconductor.	4	3	3
	OR			
5a	With energy band diagram, explain how impurity concentration affects the Fermi level in n-type and p-type semiconductors.	8	2	2
5b	Distinguish between intrinsic and extrinsic semiconductors based on energy band diagrams.	6	1	1
5c	Calculate the Fermi energy of copper given that its conduction-electron density is $8.5 \times 10^{28} \text{ m}^{-3}$.	4	3	3
6a	Define superconductivity. Explain the Meissner effect and show that superconductivity is a manifestation of perfect diamagnetism.	8	2	2
6b	Describe the two-fluid model of superconductivity and explain how it accounts for zero resistance and perfect diamagnetism.	6	1	1
6c	The critical magnetic field for a superconductor at 0 K is $6.5 \times 10^4 \text{ A/m}$. Calculate the critical field at 4 K if the critical temperature $T_c = 7.2 \text{ K}$.	4	3	3
	OR			
7a	Explain with neat sketches the two types of superconductors. Give examples for each.	8	1	1
7b	Explain the Josephson effect and discuss the construction and working of a DC SQUID with its applications.	6	2	2

7c	A superconducting magnet operates at 4.2 K and produces a field of 8 T. Estimate the critical field in A/m and discuss whether the superconductor will remain in its superconducting state if $H_{c0} = 10\text{T}$ and $T_c = 7\text{K}$.	4	3	3
8a	What are thermocouples and thermopiles? Explain their working principles and applications.	8	2	2
8b	State and explain the laws of thermoelectricity with examples.	6	1	1
8c	The Seebeck coefficient of a material is $25\ \mu\text{V/K}$, its electrical conductivity is $2 \times 10^5\ \text{S/m}$, and its thermal conductivity is $1.5\ \text{W/mK}$. Calculate the figure of merit Z at $300\ \text{K}$.	4	3	3
	OR			
9a	Describe the applications of thermoelectric materials in the automotive exhaust system, refrigeration, and space programs.	8	2	2
9b	Write a detailed note on low-, mid-, and high-temperature thermoelectric materials, highlighting their properties, examples, and uses in engineering systems.	6	1	1
9c	The Peltier coefficient of a junction is $0.004\ \text{V}$. Determine the rate of heat absorption per second when a current of $2\ \text{A}$ flows through it.	4	3	3

Signature of the Scrutinizer

Signature of the Chairman

Name

Name